SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Case Study (2 QUESTIONS)

Course Code:	ITA05	**Course Title:**	COMPUTER VISION
CO Assessed:	CO3 – Precision (BL4)	**Assessment:**	Assessment Tool 1 – Case Study
Weightage:	40% of CIA	**Total Marks:**	20 (2 Questions × 10Marks)
CO1 Statement:	*Analyze and extract image features using detection and matching techniques for effective image representation and comparison..(BL4)*		
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Section / Batch:	2025	**Date:**	

2

Code of Conduct

I ___________________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Feature Detection and Matching	Preprocessing, Edge Detection, Harris Corner Detection, Hough Transform, SIFT	1	10
Feature Detection and Matching	PCA, Image Representation, Feature Matching, Image Processing Applications	2	10
		GRAND TOTAL:	**20 Marks**

Annexure A –Case Study

1. **A remote sensing application is used to align satellite images of the same geographical region captured at different times for monitoring environmental changes. The system initially relies on edge detection to identify matching regions between the images; however, due to differences in illumination, image orientation, and background details, the matching process becomes unreliable and produces inaccurate alignment results. The development team decides to replace the edge-based approach with Harris Corner Detection to identify stable and distinctive feature points. Analyze the limitations of edge detection in this application and explain how Harris Corner Detection improves feature localization, matching accuracy, and robustness for image registration under varying imaging conditions.**

2. **An autonomous surveillance system continuously captures high-resolution color images to detect vehicles and pedestrians in real time. Processing the original color images directly increases computational complexity and slows down feature extraction, making it difficult to achieve the required processing speed. To improve system performance, the engineers consider converting the images into grayscale or binary representations before applying feature detection algorithms. Analyze the impact of different image representations on computational efficiency, edge detection, and feature extraction performance, and justify the most appropriate image representation for achieving accurate and reliable object detection in real-time computer vision applications.**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Faculty In-charge
Signature: ________________
Date: ____________________

1. Remote Sensing Image Registration Using Harris Corner Detection

Introduction

Remote sensing involves capturing images of the Earth's surface using satellites or aerial sensors. Images of the same geographical region are often captured at different times to monitor environmental changes such as deforestation, urban expansion, floods, agricultural development, and changes in land use. To compare these images accurately, the images must first be registered, meaning that corresponding points in the images must be aligned spatially.

Initially, an edge-based approach may be used to identify corresponding regions. However, edge detection alone is not sufficient for reliable image registration because edges are often affected by illumination, orientation, background changes, and noise. Harris Corner Detection provides a more suitable approach because it identifies distinctive points, known as corners, which can serve as reliable feature points for image matching and registration.

Limitations of Edge Detection

Edge detection identifies locations where there is a significant change in image intensity. Operators such as Sobel, Prewitt, and Canny can detect boundaries of roads, buildings, rivers, forests, and other geographical structures. Although these edges provide useful structural information, they have several limitations for image registration.

First, edges are not highly distinctive features. A long straight road, for example, may produce a continuous edge containing many similar points. It becomes difficult to determine which exact point on one edge corresponds to a point on the same edge in another image. This can lead to ambiguous feature matching.

Second, edge detection is sensitive to illumination variations. Satellite images acquired at different times may have different brightness and contrast because of changes in sunlight, atmospheric conditions, shadows, and sensor characteristics. These changes can alter the intensity gradients and cause some edges to become weaker or disappear.

Third, background changes can generate false or additional edges. Vegetation, clouds, shadows, water reflections, and temporary objects may appear differently in images captured at different times. The edge detector may therefore identify features that do not correspond to the same geographical structures.

Another limitation is that edges provide information mainly along a particular direction. An edge is essentially a one-dimensional structure, so it does not provide as much local positional information as a distinctive corner. Consequently, determining accurate correspondence between two images can be difficult.

Furthermore, changes in image orientation and geometric transformations can make edge correspondence unreliable. If the satellite images have different viewing angles, rotations, or small geometric distortions, the detected edges may not overlap exactly.

Harris Corner Detection

Harris Corner Detection is a feature detection technique used to identify points where the image intensity changes significantly in more than one direction. Such points are called corners and are usually more distinctive and suitable for feature matching.

The Harris detector calculates the image gradients in the horizontal and vertical directions. These gradients are used to construct the second-moment matrix:

$$
M =
\begin{bmatrix}
\sum I_x^2 & \sum I_xI_y\\
\sum I_xI_y & \sum I_y^2
\end{bmatrix}
$$

The Harris response function is given by:

$$
R = \det(M)-k[\text{trace}(M)]^2
$$

where (I_x) and (I_y) represent image gradients and (k) is an empirical constant.

A large positive value of (R) indicates a strong corner. Flat regions have little intensity variation, edges have significant variation mainly in one direction, while corners have significant variation in two directions.

Improvement in Feature Localization

Harris Corner Detection improves feature localization because it identifies specific points rather than entire boundaries. For example, the intersection of two roads or the corner of a building can be represented by a precise coordinate.

These points can act as control points during image registration. If the same building corner or road intersection is detected in images captured at different times, the coordinates of those points can be compared to determine the transformation required to align the images.

Thus, Harris corners provide more precise spatial information than long, continuous edges.

Improvement in Matching Accuracy

Corners are generally more distinctive than ordinary edge points because the intensity changes around a corner occur in multiple directions. This makes the local neighborhood around a corner easier to distinguish from other image regions.

For example, a road intersection may appear at only a few locations in an image. If the same intersection is detected in another satellite image, it can provide a strong correspondence between the two images.

As a result, Harris features can reduce ambiguous matches and improve the accuracy of estimating the transformation between images.

Robustness Under Varying Imaging Conditions

Remote sensing images can be affected by changes in illumination, contrast, shadows, atmospheric conditions, and background objects. Harris Corner Detection is generally more reliable than simple edge detection because it selects strong local structures where intensity changes occur in multiple directions.

Moderate changes in brightness or contrast may change the intensity values but can still preserve strong structural features such as building corners and road intersections. Therefore, these features can remain useful for registration even when the images are not identical.

However, basic Harris Corner Detection is not completely invariant to large rotations, scale changes, or severe illumination changes. In practical applications, it may be combined with suitable feature descriptors and matching algorithms to obtain greater robustness.

Role in Image Registration

After detecting Harris corners in both satellite images, corresponding feature points can be matched. These matched points can then be used to estimate a geometric transformation such as:

- Translation
- Rotation
- Scaling
- Affine transformation
- Perspective transformation, when appropriate

The estimated transformation is then applied to one image so that it aligns with the reference image.

Overall Comparison

Feature	Edge Detection	Harris Corner Detection
Feature type	Boundaries/edges	Distinctive corner points
Localization	Less precise	More precise
Distinctiveness	Often low	Generally high
Matching	Can be ambiguous	More reliable
Illumination changes	Can affect detected edges	More stable for strong structural corners

Feature	Edge Detection	Harris Corner Detection
Background changes	May produce unwanted edges	Focuses on distinctive local structures
Registration suitability	Limited when used alone	Better for point-based registration
Geometric transformation estimation	Less reliable	More suitable using matched points

Conclusion

Therefore, edge detection alone is not an ideal feature extraction method for remote sensing image registration because edges are often continuous, non-distinctive, and sensitive to illumination, background changes, and geometric variations. Harris Corner Detection identifies stable and distinctive points where intensity changes occur in multiple directions. These points provide accurate feature localization and more reliable correspondence between images. Consequently, Harris Corner Detection improves the estimation of geometric transformations and produces more accurate image registration for applications such as environmental monitoring and satellite image change detection.

2. Image Representation for Real-Time Object Detection

Introduction

An autonomous surveillance system continuously captures high-resolution images to detect objects such as vehicles and pedestrians. Since the system must operate in real time, it must process a large number of image frames within a short period. Processing high-resolution color images directly requires considerable computational resources because each pixel contains information from three color channels.

Therefore, the system can use different image representations, such as RGB color, grayscale, and binary images, before performing edge detection and feature extraction. Each representation provides a different balance between computational efficiency and information preservation.

RGB Color Representation

A conventional RGB image consists of three channels: red, green, and blue. Each pixel contains three intensity values. Therefore, a color image contains approximately three times as much channel data as a single-channel grayscale image of the same resolution.

Processing RGB images directly increases:

- Memory requirements
- Number of pixel operations

- Processing time
- Computational complexity

For example, if an image has (1920×1080) pixels, processing three color channels requires operations on approximately three times the channel data compared with a single grayscale channel.

However, RGB images preserve color information, which can be useful for distinguishing objects from their surroundings. Therefore, color should not be discarded when it is an important feature for the particular detection task.

Grayscale Representation

A grayscale image contains only one intensity channel. It represents the brightness of each pixel rather than its individual red, green, and blue components.

A common RGB-to-grayscale conversion is:

$$
Gray = 0.299R + 0.587G + 0.114B
$$

The conversion reduces the data that subsequent image-processing algorithms must handle.

This provides a significant computational advantage. Instead of performing feature extraction on three channels, the system can perform it on a single channel.

Most traditional feature-detection techniques are primarily based on intensity gradients. Therefore, color information is often unnecessary for detecting basic structural features such as edges, corners, and contours.

Effect of Grayscale on Edge Detection

Grayscale is highly suitable for edge detection because edge detectors primarily analyze changes in intensity.

For example, the boundary between a vehicle and the road may produce a strong intensity gradient. Similarly, the outline of a pedestrian can be detected from changes in brightness between the person and the background.

Algorithms such as:

- Sobel
- Prewitt
- Roberts
- Canny

can operate effectively on grayscale images.

Using grayscale also avoids unnecessarily applying the edge detector separately to all RGB channels, thereby reducing computational requirements and improving processing speed.

Binary Representation

A binary image contains only two possible values, generally 0 and 1 or black and white. It can be obtained from a grayscale image using thresholding.

The basic thresholding operation can be represented as:

$$
g(x,y)=
\begin{cases}
1, & f(x,y)>T\\
0, & f(x,y)\leq T
\end{cases}
$$

where (T) is the threshold value.

Binary representation significantly reduces the amount of information in an image. This makes some operations, such as connected-component analysis, contour detection, and simple shape analysis, computationally efficient.

However, this reduction in information is also its major disadvantage.

Limitations of Binary Images

Binary images discard most intensity information. If the threshold is not selected correctly, important portions of a vehicle or pedestrian may be removed.

For example, consider a pedestrian wearing dark clothes standing against a dark background. A simple threshold may fail to separate the pedestrian from the background.

Similarly, shadows can be incorrectly classified as objects. Changes in sunlight or artificial lighting can also cause the same object to have different pixel values in different frames.

Therefore, binary representation may provide excellent computational efficiency but can reduce detection reliability in complex real-world surveillance environments.

Effect on Feature Extraction

Feature extraction identifies important image characteristics such as:

- Edges
- Corners
- Contours
- Shapes
- Texture
- Intensity gradients

RGB images contain extensive information, but extracting features from all three channels requires greater computational effort.

Grayscale images preserve the intensity information required by many traditional feature extraction techniques while reducing the data size. Therefore, they provide a good compromise between computational efficiency and feature quality.

Binary images are useful for extracting simple shape and contour features, but they may lose important information needed to distinguish objects accurately.

Effect of Illumination

Real-time surveillance systems operate under different lighting conditions. For example, vehicles and pedestrians may be captured during bright daylight, cloudy weather, evening conditions, or under artificial street lighting.

Binary representation is particularly sensitive to illumination because small changes in intensity can cause pixels to cross the threshold. This can result in missing object regions or false detections.

Grayscale representation retains the intensity variations and therefore generally provides more information for reliable edge and feature extraction.

Comparison

Parameter	RGB	Grayscale	Binary
Number of channels	3	1	1
Computational cost	High	Low/Moderate	Very low
Information retained	Color + intensity	Intensity	Only two intensity classes
Edge detection	Good	Excellent for classical methods	Limited
Feature extraction	Rich but expensive	Efficient and effective	Mainly shape/contour

Parameter	RGB	Grayscale	Binary
Sensitivity to threshold	No	No direct threshold required	High
Robustness to lighting	Good information availability	Generally good	Relatively poor
Real-time processing	More demanding	Highly suitable	Very fast
Overall suitability	When color is important	Best general compromise	Useful after segmentation

Most Appropriate Representation

For the described surveillance system, grayscale is the most appropriate general-purpose representation when the subsequent processing relies on traditional edge detection and feature extraction.

The reason is that grayscale reduces the computational complexity from three color channels to one while preserving the intensity information needed to identify important structural features. It allows algorithms such as Canny, Sobel, Harris Corner Detection, contour extraction, and gradient-based feature extraction to operate efficiently.

Binary representation should not normally be selected as the primary representation for the entire detection pipeline because it removes too much information and is sensitive to threshold selection and illumination changes. Instead, binary images can be generated at a later stage when segmentation or foreground extraction is required.

RGB should be retained when color itself is an important discriminative feature. For example, if the surveillance system must specifically distinguish objects based on color, converting immediately to grayscale may reduce useful information.

Recommended Processing Pipeline

A suitable classical computer vision pipeline would be:

RGB Input → Grayscale Conversion → Noise Reduction → Edge/Feature Detection → Segmentation/Object Detection → Object Classification

This approach reduces unnecessary computation while retaining sufficient structural information for detecting vehicles and pedestrians.

Conclusion

RGB images contain the maximum amount of visual information but require greater computational resources. Binary images are computationally efficient but discard significant intensity information and are highly dependent on threshold selection. Grayscale images provide the best balance between computational efficiency, edge detection capability, feature extraction quality, and detection reliability.

Therefore, for a real-time surveillance system using conventional computer vision techniques, grayscale is the most appropriate primary image representation. It substantially reduces processing requirements while retaining the intensity and structural information necessary for reliable vehicle and pedestrian detection. Binary representation can subsequently be used for segmentation and shape analysis when appropriate, while RGB can be retained when color information is essential to the detection task.